

Happy Heart Test Plan

Scope: What to Test and What Not to Test

In-Scope Testing

- **Monitor Vitals & Their Thresholds:**
 - Pulse Rate
 - Blood Oxygen Level
 - Blood Pressure
- **Program Reads Input From:**
 - A file from the command line
 - Keyboard
- **Raise Alarms:**
 - Raise 3 levels of alarm (Low, medium & high)
 - Alarm persists until the problem is resolved
 - Only report the highest alarm level
 - Alarms provide short description of the problem
- **The Program Will:**
 - Process input every 10 seconds
 - Generate time-stamped status messages
 - Calculate oxygen moving averages
 - Handle impossible/invalid data

Out-of-Scope

- Performance testing
- GUI testing
- Hardware integration

Strategy:

Black Box Testing:

A. Equivalence Partitioning

- Pulse partition:
 - Valid range 0-260 (integer)
 - Invalid range: <0, >260, non- integer values, missing value

- P1 0-19 critically low pulse, Highest Alarm
- P2 20-39 very low pulse, Medium Alarm
- P3 40-110 Normal pulse
- P4 111-130 elevated pulse, Low Alarm
- P5 131-170 High pulse, Medium Alarm
- P6 171-210 Extremely high pulse, Highest Alarm
- Oxygen Level partition:

Valid range: 0-99.9 (decimal to one place)

Invalid range: ≤ 0 , ≥ 100 , non- decimal values

 - O1 < 50.0 critically low oxygen level, Highest Alarm
 - O2 50-79.9 dangerously low oxygen level, Medium Alarm
 - O3 80-84.9 slightly low oxygen level, Low Alarm
 - O4 85-99.9 normal oxygen level
- Blood Pressure partition:

Valid range: Systolic 0-230 and Diastolic 0-150

Invalid range: Systolic < 0 , > 231 and Diastolic: < 0 , > 150 , non-integer values

 - BP1 0 - 49/ 0-32 critically low blood pressure, Highest Alarm
 - BP2 50-69/ 33-39 too low blood pressure, Medium Alarm
 - BP3 70-150/ 40-90 normal blood pressure
 - BP4 151-200 / 91-120 evaluated blood pressure, Low Alarm
 - BP5 201-231/ 121-150 high blood pressure, Medium

B. Boundary Value Testing

- Pulse Boundary
 - Below minimum value : -1
 - Minimum value: 0
 - Maximum value: 260
 - Above maximum value: 261
- Oxygen Boundary
 - Below minimum value : -.1
 - Minimum value: 0.0
 - Maximum value: 99.9
 - Above maximum value: 100
- Blood Pressure Boundary
 - Below minimum value : -1/ -1
 - Minimum value: 0/0
 - Maximum value: 230/150
 - Above maximum value: 231/151

White Box Testing:

A. Statement Coverage:

- Pulse alarm levels
- Oxygen alarm levels
- Missing oxygen logic
- Moving average calculation
- Blood Pressure alarm
- Alarm precedence logic
- Input validation

B. Branch Coverage:

- If/else conditions for all thresholds
- Alarm continuation logic
- File vs keyboard input

C. Loop Coverage

- Main input loop with differing amount of lines per .txt file
- Alarm precedence loop with differing values in alarms list

Resource:

- **Software Resources**
 - Backend: Python
- **People**
 - Jonny Jackson
 - Matthew Jorgenson
 - Tyler Webster
 - Colton evans
- **Hardware**
 - Laptop Mac & Windows
- **Software**
 - Pytest for Unit testing

Timeline:

- **Milestone 1**
 - Source Code

- Test Plan
- Working executable module
- (future milestone requirements not yet available)

Risk:

- Boundary conditions not handled correctly
- Calculation errors due to data type issues
- Time constraints limiting full coverage

Tests:

Input Validation					
Test Case	Pulse	Oxygen	Blood Pressure	Data	Expected Result
TC-1	Valid	Valid	Valid	70 99.0 120/80	Valid
TC-2	Valid	Valid	Invalid	70 99.0 a	Invalid
TC-3	Valid	Invalid	Valid	70 a 120/80	Invalid
TC-4	Invalid	Valid	Valid	a 99.0 120/80	Invalid
TC-5	Valid	Invalid	Invalid	70 a a	Invalid
TC-6	Invalid	Valid	Invalid	a 99.0 a	Invalid
TC-7	Invalid	Invalid	Valid	a a 120/80	Invalid
TC-8	Invalid	Invalid	Invalid	a a a	Invalid

Input Validation - Pulse				
Test Case	Input	Partition	Alarm Level	Result

TC-9	175	Extremely High Pulse	Highest Alarm	Valid
TC-10	80	Normal Pulse	None	Valid
TC-11	32	Critically Low Pulse	Highest Alarm	Valid
TC-12	276	Invalid	Equipment Malfunction	Invalid
TC-13	b	Invalid	N/A	Invalid
TC-14	155	Elevated Pulse	Low Alarm	Valid
TC-15	-1	Invalid	Equipment Malfunction	Invalid

Input Validation - Oxygen Level				
Test Case	Input	Partition	Alarm Level	Result
TC-16	90.8	Normal Oxygen Level	None	Valid
TC-17	-3	Invalid	Equipment Malfunction	Invalid
TC-18	32.9	Very Low Oxygen Level	Medium Alarm	Valid
TC-19	150	Invalid	Equipment Malfunction	Invalid
TC-20	83.5	Slightly low oxygen	Low Alarm	Valid

TC-21	a	Invalid	N/A	Invalid
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Input Validation - Blood Pressure				
Test Case	Input	Partition	Alarm Level	Result
TC-22	40/20	Critically Low Blood Pressure	Highest Alarm	Valid
TC-23	120/80	Normal Blood Pressure	None	Valid
TC-24	66/36	Low Blood Pressure	Medium Alarm	Valid
TC-25	276/175	Invalid	Equipment Malfunction	Invalid
TC-26	155/93	Evaluated Blood Pressure	Low Alarm	Valid
TC-27	a/t	Invalid	N/A	Invalid

Boundary Value - Pulse			
Test Case	Input	Boundary	Result
TC-28	278	Above	Invalid

		Maximum Value	
TC-29	-34	Below Minimum Value	Invalid
TC-30	260	Maximum Value	Valid
TC-31	0	Minimum Value	Valid

Boundary Value - Oxygen Level			
Test Case	Input	Boundary	Result
TC-32	-0.1	Below Minimum Value	Invalid
TC-33	0.0	Below Minimum Value	Valid
TC-34	99.9	Maximum Value	Valid
TC-35	100	Above Maximum Value	Invalid

Boundary Value - Blood Pressure			
Test	Input	Boundary	Result

Case			
TC-36	-1/-1	Below Minimum Value	Invalid
TC-37	0/0	Minimum Value	Valid
TC-38	260/150	Maximum Value	Valid
TC-39	261/151	Above Maximum Value	Invalid

Test that alarm levels are prioritized correctly

Test Case	Test Title	Input	Expected Result
TC-40	High pulse vs <i>Medium</i> Oxygen	10 75 120/80	00:00 Alarm.HIGHEST Alarm Pulse critically low (10)
TC-41	Oxygen <i>High</i> vs Pulse <i>Medium</i>	35 40 120/80	00:00 Alarm.HIGHEST Alarm Oxygen critically low (40.0%)
TC-42	Blood Pressure <i>High</i> vs Pulse <i>Low</i>	115 92 40/30	00:00 Alarm.HIGHEST Alarm Blood pressure critically low (40/30)
TC-45	Pulse <i>Medium</i> vs Oxygen <i>Medium</i>	35 75 120/80	00:00 Alarm.MEDIUM Alarm Pulse very low (35)
TC-46	Oxygen <i>Medium</i> vs Blood Pressure	80 75 201/121	00:00 Alarm.MEDIUM Alarm Oxygen dangerously low (75.0%)

	<i>Medium</i>		
TC-47	Pulse <i>Low</i> vs Oxygen <i>Low</i>	115 84 120/80	00:00 Alarm.LOW Alarm Pulse elevated (115)
TC-48	Oxygen <i>Low</i> Vs Blood Pressure <i>Low</i>	80 84 151/91	00:00 Alarm.LOW Alarm Oxygen slightly low (84.0%)
TC-49	New Pulse <i>High</i>	80 92 201/121 10 92	00:10 Alarm.HIGHEST Alarm Pulse critically low (10)
TC-50	New Oxygen <i>Medium</i>	80 92 201/121 80 75	00:10 Alarm.MEDIUM Alarm Oxygen dangerously low (75.0%)

Test rolling average for oxygen levels

Test Case	Test Title	Input	Expected Result
TC-51	Single Reading Avg	80 90	00:00 None Everything normal
TC-52	Double reading Avg	80 90 80 80	00:10 None Everything normal
TC-53	Triple Reading Avg (Low Alarm)	80 90 80 80 80 80	00:20 Alarm.LOW Alarm Oxygen slightly low (83.3%)
TC-54	Medium Threshold	80 75 80 75 80 75	00:20 Alarm.MEDIUM Alarm Oxygen dangerously low (75.0%)
TC-55	High Threshold	80 45 80 45 80 45	00:20 Alarm.HIGHEST Alarm Oxygen critically low (45.0%)

TC-56	Oxygen Minute Avg	80 90 80 90 80 90 80 90 80 90 80 60 80 60	01:00 Alarm.LOW Alarm Oxygen slightly low (80.0%)
TC-57	Reuse Missing Value	80 80 80	00:10 Alarm.LOW Alarm Oxygen slightly low (80.0%)
TC-58	Reuse 2 Missing Values	80 80 80 80	00:20 Alarm.LOW Alarm Oxygen slightly low (80.0%)
TC-59	Reuse 3 Missing Values	80 80 80 80 80	00:30 Alarm.LOW Alarm Oxygen sensor missing
TC-60	Relocating Oxygen Reading	80 80 80 80 80 80 80	00:40 Alarm.LOW Alarm Oxygen slightly low (80.0%)